MA 102 (Mathematics II - Ordinary Differential Equations)

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Solutions - Tutorial Sheet No. 3 Date of Discussion: March 27, 2024

Lectures 7-10: Higher-order differential equations, Introduction to power series

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Attention students: Please keep in mind that it is not possible to discuss all problems during the tutorial. You can consider the problems not discussed as your practice problems.

1. Consider the differential equation $y'' - 5y' + 6y = 0$.

   (i) Show that e^{2x} and e^{3x} are linearly independent solutions of this equation in $-\infty < x < \infty$.

   (ii) Write the general solution of the given equation.

   (iii) Find the solution that satisfies the conditions $y(0) = 2$, $y'(0) = 3$. Is this solution unique? Justify your answer.

   Solution: (i) It is quite straightforward to observe that both e^{2x} and e^{3x} satisfy the given differential equation. Next,

$$\begin{aligned} W(e^{2x}, e^{3x})(x) &= \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix} \\ &= 3e^{5x} - 2e^{5x} \\ &\neq 0 \text{ for any } x \in \mathbb{R}. \end{aligned}$$

   It implies that e^{2x} and e^{3x} are linearly independent solutions.
   (ii) Therefore, the general solution can be written as

$$y(x) = c_1 e^{2x} + c_2 e^{3x}.$$

   (iii) Using the conditions $y(0) = 2,\ y'(0) = 3$, we get the following equations in c_1 and c_2:

$$\begin{aligned} c_1 + c_2 &= 2 \\ 2c_1 + 3c_2 &= 3 \end{aligned}$$

   Solving them, we get $c_1 = 3$ and $c_2 = -1$. Hence, $y(x) = 3e^{2x} - e^{3x}$ is the unique solution satisfying the ODE and the given initial conditions.

2. If y_1 and y_2 are linearly independent solutions of $xy'' + 2y' + xe^x y = 0$, $0 < x < \infty$ and if $W(y_1, y_2)(1) = 2$, find the value of $W(y_1, y_2)(5)$.
   Solution: Abel's formula is

$$W(y_1, y_2)(x) = W(y_1, y_2)(x_0) \exp\left(-\int_{x_0}^{x} P(\xi)d\xi\right).$$

Using the above,

$$
\begin{aligned}
W(y_1,y_2)(5) &= W(y_1,y_2)(1)\exp\left(-\int_1^5 P(\xi)d\xi\right)\\
&= 2\exp\left(-\int_1^5 (2/\xi)d\xi\right)\\
&= 2\exp(-2\ln 5)\\
&= 2\exp(\ln(1/25))\\
&= 2/25.
\end{aligned}
$$

3. Let $P(x), Q(x) \in C(I)$. Assume that the functions $y_1,\ y_2 \in C^2(I)$ are solutions of the differential equations $y'' + P(x)y' + Q(x)y = 0$ on an open interval I. Prove that
(a) if y_1 and y_2 are zero at the same point in I, then they cannot be a fundamental set of solutions on that interval;
(b) if y_1 and y_2 have a common point of inflection x_0 in I, then they cannot be a fundamental set of solutions on that interval.
Solution:
(a) Let us take $y_1(x_0) = 0,\quad y_2(x_0) = 0$. Then

$$
\begin{aligned}
W(y_1,y_2)(x_0) &= \begin{vmatrix} y_1(x_0) & y_2(x_0)\\ y_1'(x_0) & y_2'(x_0)\end{vmatrix}\\
&= \begin{vmatrix} 0 & 0\\ y_1'(x_0) & y_2'(x_0)\end{vmatrix}\\
&= 0.
\end{aligned}
$$

Therefore, y_1 and y_2 are not linearly independent solutions of the given ODE implying that they cannot be the fundamental set of solutions on I containing a common zero x_0.
(b) Let y_1 and y_2 have a common inflexion point x_0 in I, that is, there exists $x_0 \in I$ such that

$$y_1''(x_0) = 0,\quad y_2''(x_0) = 0.$$

With the above and also that y_1 and y_2 are solutions of the ODE, we have the following:

$$
\begin{aligned}
P(x_0)y_1'(x_0) + Q(x_0)y_1(x_0) = 0,\\
P(x_0)y_2'(x_0) + Q(x_0)y_2(x_0) = 0.
\end{aligned}
$$

This can be written as

$$\begin{bmatrix} y_1'(x_0) & y_1(x_0)\\ y_2'(x_0) & y_2(x_0)\end{bmatrix}\begin{bmatrix} P(x_0)\\ Q(x_0)\end{bmatrix} = \begin{bmatrix} 0\\ 0\end{bmatrix}. \tag{1}$$

Let us assume that $(P(x_0), Q(x_0)) \neq (0,0)$ so that P and Q do not vanish simultaneously. Then, (1) holds only when $\det\begin{bmatrix} y_1(x_0) & y_2(x_0)\\ y_1'(x_0) & y_2'(x_0)\end{bmatrix}$ is zero. This in turn means that $W(y_1,y_2)(x_0) = 0$. We can now conclude that y_1 and y_2 cannot be linearly independent solutions and hence cannot form the fundamental set of solutions.

4. Find the general solution of the following differential equations:

(a) $\dfrac{d^4y}{dx^4} + y = 0$

(b) $\dfrac{d^5y}{dx^5} - 2\dfrac{d^4y}{dx^4} + \dfrac{d^3y}{dx^3} = 0$

(c) $\dfrac{d^3y}{dx^3} - \dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} - y = 0$

(d) $\dfrac{d^5y}{dx^5} + 5\dfrac{d^4y}{dx^4} + 10\dfrac{d^3y}{dx^3} + 10\dfrac{d^2y}{dx^2} + 5\dfrac{dy}{dx} + y = 0$

Solution: All given ODEs are linear homogeneous ODEs with constant coefficients.
(a) The auxiliary equation is $m^4 + 1 = 0$. We know that the n-th roots of $z = r(\cos\theta + \mathrm{i}\sin\theta)$ are given by

$$z^{1/n} = r^{1/n}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + \mathrm{i}\sin\left(\frac{\theta + 2k\pi}{n}\right)\right], \quad k = 0, 1, \ldots, n-1.$$

For $n = 4$, we have

$$z^{1/4} = \left[\cos\left(\frac{\pi + 2k\pi}{4}\right) + \mathrm{i}\sin\left(\frac{\pi + 2k\pi}{4}\right)\right], \quad k = 0, 1, \ldots, n-1.$$

The roots can be obtained as

$$\frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}\mathrm{i}, \quad \frac{-\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}\mathrm{i}$$

. The general solution is given by

$$y(x) = e^{\frac{x}{\sqrt{2}}}\left[c_1 \sin\frac{x}{\sqrt{2}} + c_2 \cos\frac{x}{\sqrt{2}}\right] + e^{-\frac{x}{\sqrt{2}}}\left[c_3 \sin\frac{x}{\sqrt{2}} + c_4 \cos\frac{x}{\sqrt{2}}\right]$$

(b) Here, the auxiliary equation is $m^5 - 2m^4 + m^3 = 0$ giving $m = 0$ as a root of multiplicity 3 and $m = 1$ as a root of multiplicity 2. Hence, the general solution is

$$y(x) = (c_1 + c_2x + c_2x^2) + (c_4 + c_5x)e^x.$$

(c) The auxiliary equation is $m^3 - m^2 + m - 1 = 0$ which can be written as $(m^2+1)(m-1) = 0$ giving the roots as $m = 1, \ \pm\mathrm{i}$. The general solution is

$$y(x) = c_1e^x + c_2\cos x + c_3\sin x.$$

(d) The auxiliary equation is $m^5 + 5m^4 + 10m^3 + 10m^2 + 5m + 1 = 0$ which can be written as $(m+1)^5 = 0$ giving $m = -1$ as a root of multiplicity 5. The general solution is

$$y(x) = (c_1 + c_2x + c_3x^2 + c_4x^3 + c_5x^4)e^{-x}.$$

5. Solve the following initial-value problems:

(a) $y'' - 2y' + y = 2xe^{2x} + 6e^x$; $y(0) = 1, \ y'(0) = 0$
(b) $y''(x) + y(x) = 3x^2 - 4\sin x, \ y(0) = 0, \ y'(0) = 1$
Solution:
(a) The complementary function can be obtained as $y_c = (c_1 + c_2x)e^x$ whereas the

particular integral can be tried as $y_p = Axe^{2x} + Be^{2x} + Cx^2e^x$ and obtained as $y_p = 2xe^{2x} - 4e^{2x} + 3x^2e^x$. The general solution takes the form

$$y(x) = (c_1 + c_2x)e^x + 2xe^{2x} - 4e^{2x} + 3x^2e^x.$$

The initial conditions are given as $y(0) = 1,\ \ y^{'}(0) = 0$. Hence, we get $c_1 = 5$ and $c_2 = 1$. Hence, the solution can be written as

$$y(x) = (x + 5)e^x + 3x^2e^x + 2xe^{2x} - 4e^{2x}.$$

(b) The complementary function can be obtained as $y_c = c_1 \cos x + c_2 \sin x$. For finding the particular integral, try $y_p = Ax^2 + Bx + C + Dx \cos x + Ex \sin x$. We will get $A = 3,\ B = 0,\ C = -6,\ D = 2,\ E = 0.$. The particular integral can be obtained as $y_p = 3x^2 - 6 + 2x \cos x$. The general solution takes the form

$$y(x) = c_1 \cos x + c_2 \sin x + 3x^2 - 6 + 2x \cos x.$$

The initial conditions are given as $y(0) = 0,\ \ y^{'}(0) = 1$. Hence, we get $c_1 = 6$ and $c_2 = -1$. Hence, the solution can be written as

$$y(x) = 6 \cos x - \sin x + 3x^2 - 6 + 2x \cos x.$$

6. Use the method of undermined coefficients to find a particular solution to the following differential equations:

   (a) $y'' - 3y' + 2y = 2x^2 + 3e^{2x}$
   (b) $y'' - 3y' + 2y = xe^{2x} + \sin x$

   Solution:
   (a) Try $y_p = Ax^2 + Bx + C + Dxe^{2x}$. Putting $y, y', y^{''}$ in the given equation, we can obtain $A = 1,\ B = 3,\ C = 7/2\ D = 3$. Hence, the particular integral is $y_p = x^2 + 3x + 3xe^{2x} + 7/2$.
   (b) Try $y_p = Ax^2e^{2x} + Bxe^{2x} + C \sin x + D \cos x$. Putting $y, y', y^{''}$ in the given equation, we can obtain $A = 1/2,\ B = -1,\ C = 1/10\ D = 3/10$. Hence the particular integral is

$$y_p = \frac{1}{2}x^2e^{2x} - xe^{2x} + \frac{1}{10}\sin x + \frac{3}{10}\cos x.$$

7. Use the method of variation of parameters to find a particular solution, and hence the general solution for the following differential equations:

   (a) $y'' - 5y' + 6y = \cos(2x) + 1$
   (b) $y'' - 5y' + 6y = e^{3x} - x^2$
   Solution:
   (a) The complementary function cane be found as $y_c = Ae^{2x} + Be^{3x}$. Hence, the complete solution can be written as

$$y_c = A(x)e^{2x} + B(x)e^{3x}. \tag{2}$$

   The following equations arise for the unknowns A and B:

$$\begin{aligned} A'(x)e^{2x} + B'(x)e^{3x} &= 0 \\ 2A'(x)e^{2x} + 3B'(x)e^{3x} &= \cos 2x + 1, \end{aligned}$$

from which $A'(x)$ and $B'(x)$ can be obtained as

$$A'(x) = -(1+\cos 2x)e^{-2x}, \quad B'(x) = (1+\cos 2x)e^{-3x}.$$

After integrating, we find

$$\begin{aligned} A(x) &= \frac{e^{-2x}}{2} - \frac{e^{-2x}(-2\cos 2x + 2\sin 2x)}{8} + c_1 \\ B(x) &= -\frac{e^{-3x}}{3} + \frac{e^{-3x}(-3\cos 2x + 2\sin 2x)}{13} + c_2. \end{aligned}$$

The following integration formula has been used:

$$\int e^{ax}\cos bx\, dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}.$$

Inserting these expressions of $A(x)$ and $B(x)$ in (2) and taking away the complementary functions $c_1e^{2x} + c_2e^{3x}$, the exact form of the particular integral can be obtained.

(b) The complementary function is the same as the above problem (refer to equation (2)). The following equations arise for the unknowns A and B:

$$\begin{aligned} A'(x)e^{2x} + B'(x)e^{3x} &= 0 \\ 2A'(x)e^{2x} + 3B'(x)e^{3x} &= e^{3x} - x^2. \end{aligned}$$

$A'(x)$ and $B'(x)$ can be obtained as

$$A'(x) = x^2e^{-2x} - e^x, \quad B'(x) = 1 - x^2e^{-3x}.$$

After integrating, we find

$$\begin{aligned} A(x) &= -\frac{1}{2}x^2e^{-2x} - \frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} - e^x + c_1 \\ B(x) &= x + \frac{1}{3}x^2e^{-3x} + \frac{2}{9}xe^{-3x} + \frac{2}{27}e^{-3x} + c_2. \end{aligned}$$

Inserting these expressions of $A(x)$ and $B(x)$ in (2) and taking away the complementary functions $c_1e^{2x} + c_2e^{3x}$, the exact form of the particular integral can be obtained.

8. Use the operator method to find a particular solution of the following ODEs:
(a) $y''' + y'' + y' + y = x^5 - 2x^2 + x$
(b) $y''' - 5y'' + 8y' - 4y = 3e^{2x}$
(c) $y'' - 3y' + 2y = 3\sin 2x$
Solution:
(a) The given equation can be written as

$$(D^3 + D^2 + D + 1)y = x^5 - 2x^2 + x.$$

The particular integral can be found as

$$\begin{aligned} y_p &= \frac{x^5 - 2x^2 + x}{D^3 + D^2 + D + 1} \\ &= \frac{1-D}{1-D^4}(x^5 - 2x^2 + x) \\ &= \frac{1}{1-D^4}(x^5 - 5x^4 - 2x^2 + 5x - 1) \\ &= (1 + D^4 + D^8 + \cdots)(x^5 - 5x^4 - 2x^2 + 5x - 1) \\ &= x^5 - 5x^4 - 2x^2 + 125x - 121 \end{aligned}$$

(b) The given equation can be written as

$$((D-1)^2(D-1))y = 3e^{2x}.$$

The particular integral can be found as

$$\begin{aligned} y_p &= \frac{3e^{2x}}{(D-2)^2(D-1)} \\ &= 3e^{2x}\frac{1}{D^2(D+1)} \\ &= \cdots \\ &= \frac{3x^2e^{2x}}{2} \end{aligned}$$

(c) The given equation can be written as

$$(D^2-3D+2)y = 3\sin 2x.$$

The particular integral can be found as

$$\begin{aligned} y_p &= \frac{3\sin 2x}{D^2-3D+2} \\ &= -3\frac{\sin 2x}{2+3D} \\ &= = -3\frac{(3D-2)\sin 2x}{9D^2-4} \\ &= \cdots \\ &= \frac{3}{20}\left(3\cos 2x - \sin 2x\right) \end{aligned}$$

9. If p is not zero or a positive integer, show that the series

$$\sum_{n=1}^{\infty}\frac{p(p-1)(p-2)\cdots(p-n+1)}{n!}x^n$$

converges for $|x|<1$ and diverges for $|x|>1$.
Solution: We have

$$\begin{aligned} L = \left|\frac{a_{n+1}}{a_n}\right| &= \left|\frac{p(p-1)(p-2)\cdots(p-n+1)(p-n)}{p(p-1)(p-2)\cdots(p-n+1)(n+1)}\frac{x^{n+1}}{x^n}\right| \\ &= \left|\frac{(p-n)}{n+1}x\right| \end{aligned}$$

As $n\to\infty$, $L=|x|$. Hence the series converges for $|x|<1$ and diverges for $|x|>1$.

10. We know that

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$$

is valid for $|x|<1$. Applying it, show that
(a) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$
(b) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$

for $|x| < 1$.

Solution:

(a) Given

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots . \tag{3}$$

Integrate both sides of (3) to get

$$\int \frac{dx}{1+x} = \int (1 - x + x^2 - x^3 + \cdots)\, dx$$
$$\Rightarrow \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

(b) From (3), we have

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots . \tag{4}$$

Integrating (4), we get

$$\int \frac{dx}{1+x^2} = \int (1 - x^2 + x^4 - x^6 + \cdots)\, dx$$
$$\Rightarrow \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

11. Consider the differential equation $y' + y = 1$. Find its power series solution and recognize it as the expansion of a familiar function. Further, verify your power series solution by solving the ODE by an elementary method.

Solution:

Let us assume the solution of $y' + y = 1$ in the form

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots . \tag{5}$$

Differentiating (5), we get

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots . \tag{6}$$

Putting the values of y and y' from equations (5) and (6) in the given differential equation, we get)

$$(a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots) + (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots = 1.$$

Equating various powers of x, we get

$$a_1 = -(a_0 - 1),\ \ 2a_2 = -a_1 \Rightarrow a_2 = -a_1/2 = (a_0 - 1)/2,\ \ 3a_3 = -a_2 \Rightarrow a_3 = -a_2/3 = -(a_0 - 1)/3!,$$

and so on. Putting these back in (5), we get

$$y = a_0 - (a_0 - 1)x + \frac{a_0 - 1}{2!}x^2 - \frac{a_0 - 1}{3!}x^3 + \cdots .$$

It can be written as

$$y = 1 + (a_0 - 1)\left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right).$$

Realizing the quantity in the brackets as e^{-x}, we can write the solution as

$$y(x) = 1 + (a_0 - 1)e^{-x}.$$

12. Repeat the above for the ODE $y' = x - y$ with $y(0) = 0$.
Solution:
Let us assume the solution of $y' = x - y$ in the form

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots . \tag{7}$$

Differentiating (7), we get

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots . \tag{8}$$

Putting the values of y and y' from equations (7) and (8) in the given differential equation, we get

$$a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots = x - (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots)$$

Equating various powers of x from both sides, we get

$$a_1 = -a_0, \;\; 2a_2 = 1 - a_1 \Rightarrow a_2 = -(a_1 - 1)/2 = (a_0 + 1)/2, \;\; 3a_3 = -a_2 \Rightarrow a_3 = -a_2/3 = -(a_0 + 1)/3!,$$

and so on. Putting these back in (7), we get

$$y = a_0 - a_0 x + \frac{a_0 + 1}{2!} x^2 - \frac{a_0 + 1}{3!} x^3 + \cdots ,$$

giving us

$$y = a_0 - a_0 x + (a_0 + 1)\left(\frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right).$$

Using $y(0) = 0$, we get $a_0 = 0$. Then,

$$y = \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots .$$

Now,we can write the solution in terms of an elementary function as

$$y(x) = x - 1 + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right) = x - 1 + e^{-x}.$$

The End